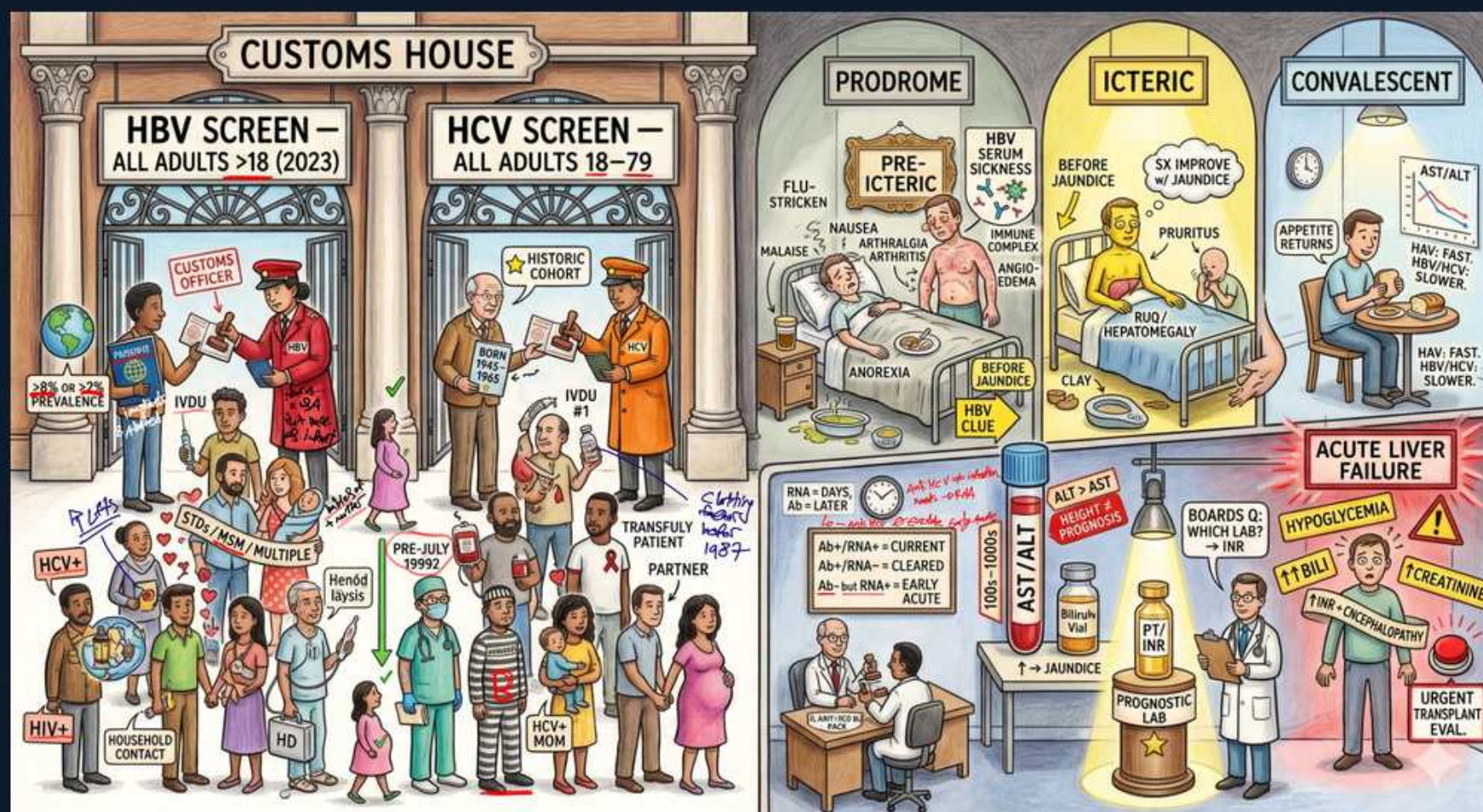


Viral Hepatitis — Acute Course, Screening & Prognostic Factors



1. HBV screen — all adults >18

WHAT YOU SEE

Left 'Customs House' arch: 'HBV SCREEN — ALL ADULTS >18 (2023)'

WHAT IT ENCODES

CDC 2023 recommends one-time hepatitis B screening for ALL adults aged 18 and older, using the triple panel: HBsAg, anti-HBs, and total anti-HBc. This replaced older risk-based-only screening. Pregnant women are screened with HBsAg at every pregnancy regardless of prior testing. Periodic re-screening is for those with ongoing risk (IVDU, HIV, multiple partners, household/sexual contacts of HBV).

BOARD TRAP

Don't pick 'screen only high-risk groups' — that is the outdated approach. Current guidance is universal one-time adult screening for HBV with the 3-test panel, not HBsAg alone.

2. HCV screen — all adults 18–79

WHAT YOU SEE

Right arch: 'HCV SCREEN — ALL ADULTS 18–79', 'HISTORIC COHORT' man

WHAT IT ENCODES

USPSTF/CDC recommend one-time hepatitis C screening with anti-HCV antibody for ALL adults aged 18–79, plus screening in pregnancy and ongoing testing for those with continued risk. The historic-cohort man captures the legacy 1945–1965 'baby boomer' birth-cohort recommendation, now folded into universal adult screening. A reactive anti-HCV requires confirmatory HCV RNA to distinguish active from cleared/past infection.

BOARD TRAP

A positive anti-HCV alone does NOT mean active infection — ~25% spontaneously clear. You must confirm with HCV RNA before diagnosing chronic HCV or treating.

3. Sexual / multiple partners

WHAT YOU SEE

'SEXUAL — MULTIPLE' couple with hearts

WHAT IT ENCODES

HBV is efficiently transmitted sexually and is the most sexually-transmissible of the hepatitis viruses; multiple partners is a major risk factor. HBV is ~50–100x more infectious than HIV per exposure. This is why HBV vaccination and screening are emphasized in those with multiple sexual partners or STIs.

BOARD TRAP

HCV is poorly transmitted sexually in monogamous heterosexual couples — high sexual transmission points to HBV (or HCV in HIV+ MSM), not routine HCV.

4. IVDU — both HBV and HCV

WHAT YOU SEE

'IVDU' figures under both arches, 'IVDU #1' for HCV

WHAT IT ENCODES

Injection drug use is a shared risk factor: it is the LEADING risk factor for HCV in the developed world ('IVDU #1') and a significant route for HBV. Needle-sharing transmits blood-borne virus efficiently. Anyone with current or past IVDU warrants screening for HBV, HCV, and HIV.

BOARD TRAP

For HCV the single most common transmission route now is IVDU — not transfusion (which is now rare due to blood-supply screening since ~1992) and not perinatal.

5. Transfusion before 1992 / blood products

WHAT YOU SEE

'TRANSFUSE PATIENT', 'PRE-JULY 1992?'

WHAT IT ENCODES

Blood transfusion before universal anti-HCV donor screening (US ~July 1992) is a classic HCV risk factor — the older patient who got a transfusion or clotting factors decades ago. After 1992, transfusion-transmitted HCV became extremely rare. Hemophiliacs treated before factor-concentrate viral inactivation are a high-risk historic cohort.

BOARD TRAP

A transfusion in 2010 is essentially zero HCV risk — the board's transfusion clue points to HCV only when it predates ~1992.

6. Perinatal / HBV mom

WHAT YOU SEE

'HCV's MOM' pregnant figure; perinatal transmission

WHAT IT ENCODES

Perinatal (vertical) transmission is the dominant global route for HBV, especially in endemic Asia/Africa — risk is highest when mother is HBeAg-positive/high viral load. Infants of HBsAg+ mothers get HBIG + birth-dose vaccine. Perinatal HCV transmission occurs but is much less efficient (~5%), higher with maternal HIV co-infection.

BOARD TRAP

Vertical transmission causing chronic infection is mainly an HBV story (neonatal infection → ~90% chronicity). HCV vertical transmission is uncommon and there is no HCV vaccine or neonatal prophylaxis.

7. Prodrome / pre-icteric

WHAT YOU SEE

'PRODROME / PRE-ICTERIC': flu-like malaise, anorexia, nausea, arthralgia, serum-sickness

WHAT IT ENCODES

The pre-icteric prodrome of acute viral hepatitis: 1–2 weeks of malaise, fatigue, anorexia, nausea, low-grade fever, and right-upper-quadrant discomfort. HBV characteristically can cause a serum-sickness-like syndrome (arthralgias, urticaria/rash) from circulating immune complexes BEFORE jaundice. RUQ pain and aversion to smoking/food are classic.

BOARD TRAP

Serum-sickness (arthritis + urticaria) preceding jaundice is a hallmark of acute HBV (immune-complex mediated) — not typically HAV or HCV.

8. Icteric phase

WHAT YOU SEE

'ICTERIC': jaundice, dark urine, pruritus, clay stool, RUQ/hepatomegaly

WHAT IT ENCODES

As bilirubin rises the patient becomes jaundiced with dark urine (conjugated bilirubinuria), pale/clay-colored stool, pruritus, and tender hepatomegaly. ALT/AST are markedly elevated (often >1000, ALT>AST in viral hepatitis). The icteric phase lasts days to weeks; in many adults (especially HCV and many HBV) infection is anicteric/subclinical.

BOARD TRAP

In viral hepatitis ALT > AST. An AST:ALT ratio >2:1 points instead to ALCOHOLIC hepatitis, not acute viral hepatitis.

9. Convalescent phase

WHAT YOU SEE

'CONVALESCENT': appetite returns, downtrending LFT curve

WHAT IT ENCODES

Recovery phase: jaundice resolves, appetite and energy return, and aminotransferases trend down toward normal over weeks. The downsloping LFT curve marks resolution. Most acute HAV and adult HBV resolve fully; the key prognostic branch points are progression to chronicity (HBV/HCV) or, rarely, to fulminant failure.

BOARD TRAP

Symptom resolution does NOT equal viral clearance in HBV/HCV — you must document HBsAg loss / undetectable HCV RNA. Clinical recovery alone can hide progression to chronic infection.

10. Serology timing — anti-HBc

WHAT YOU SEE

'RNA = DAYS LATER', 'Ab+/RNA+ = CURRENT', 'Ab+/RNA- = CLEARED', 'Ab- but RNA+ = EARLY/ACUTE'

WHAT IT ENCODES

Diagnostic windows: in acute HBV, IgM anti-HBc is the marker during the window period when HBsAg may have fallen and anti-HBs not yet risen. For HCV, antibody appears weeks after exposure, so in very early/acute HCV the Ab can be NEGATIVE while RNA is already POSITIVE (Ab-/RNA+ = early acute). Ab+/RNA+ = current infection; Ab+/RNA- = cleared/resolved.

BOARD TRAP

In suspected acute HCV, a negative anti-HCV antibody does not rule it out — order HCV RNA, which turns positive within ~1–2 weeks, well before antibody.

11. >20,000 IU/mL threshold

WHAT YOU SEE

'>20,000' vial near jaundice gauge

WHAT IT ENCODES

HBV DNA level guides chronic-HBV treatment decisions: a DNA >20,000 IU/mL (HBeAg+) or >2,000 IU/mL (HBeAg-) with elevated ALT and/or fibrosis generally warrants antiviral therapy. High viral load also predicts long-term cirrhosis/HCC risk (REVEAL study) and perinatal transmission risk (treat mothers >200,000 IU/mL in third trimester).

BOARD TRAP

Don't treat every HBV-DNA-positive patient — the immune-tolerant phase (very high DNA but normal ALT, no fibrosis) is typically monitored, not treated. Treatment hinges on DNA level + ALT + fibrosis together.

12. Prognostic lab — PT/INR

WHAT YOU SEE

'PROGNOSTIC LAB', 'PT/INR' bottle, 'BOARDS Q: WHICH LABEL = INR'

WHAT IT ENCODES

The single best prognostic test in acute hepatitis/liver failure is the PROTHROMBIN TIME / INR — it reflects hepatic synthetic function (clotting-factor production) and rises early as the liver fails. A prolonged INR (≥ 1.5) plus encephalopathy defines acute liver failure. INR, not transaminase level, predicts outcome.

BOARD TRAP

The absolute height of ALT/AST does NOT predict prognosis — falling transaminases with a RISING INR/bilirubin and new encephalopathy signals massive hepatocyte loss (worsening), not improvement.

13. Acute liver failure — King's criteria markers

WHAT YOU SEE

'ACUTE LIVER FAILURE': hypoglycemia, T.BILI, creatinine, encephalopathy, 'URGENT TRANSPLANT EVAL'

WHAT IT ENCODES

Fulminant hepatic failure = encephalopathy + coagulopathy (INR ≥ 1.5) within ~8 weeks of onset, no prior liver disease. Red flags: hypoglycemia, rising bilirubin and creatinine, lactic acidosis, and worsening encephalopathy/cerebral edema. These feed the King's College criteria that trigger urgent transplant evaluation. HBV (esp. with HDV superinfection) and HAV/HEV can cause this.

BOARD TRAP

HEV in PREGNANT women carries a strikingly high fulminant-failure mortality (~20%); and acetaminophen is the leading drug cause — King's criteria differ for acetaminophen (pH, lactate) vs non-acetaminophen (INR, bilirubin, age) failure.

14. Creatinine — hepatorenal

WHAT YOU SEE

'CREATININE' marker in liver-failure panel

WHAT IT ENCODES

Rising creatinine in acute liver failure signals hepatorenal syndrome or acute tubular necrosis and is a poor-prognostic marker incorporated into MELD and King's (non-acetaminophen) criteria. Renal dysfunction markedly worsens survival and strengthens the case for transplant listing.

BOARD TRAP

New AKI in fulminant hepatitis is an ominous prognostic sign, not an incidental finding — it raises transplant urgency rather than being managed expectantly.